

Gemini Embedding 2 Controlled Comparison

Nomic local vs Gemini Embedding 001 vs Gemini Embedding 2

Official bilingual retrieval benchmark and fixed-GLM downstream diagnostic

Executive Summary

Only Gemini Embedding 2 was executed anew. The Nomic and Gemini Embedding 001 arms were reused from frozen evidence. The official benchmark contains 60 retrieval questions; the six-case downstream QA diagnostic holds GLM 5.3 Flash fixed.

Official ranking preference: **gemini-embedding-2**

Strict superiority gate passed: **False**

Official 60-Question Results

Arm	Method	Top-1	Top-3	MRR@10	Recall@10
nomic	lexical	43.3%	71.7%	0.5809	81.7%
nomic	semantic	35.0%	65.0%	0.5061	71.7%
nomic	hybrid	63.3%	78.3%	0.7322	86.7%
gemini_001	lexical	43.3%	71.7%	0.5809	81.7%
gemini_001	semantic	71.7%	93.3%	0.8264	95.0%
gemini_001	hybrid	71.7%	95.0%	0.8306	95.0%
gemini_2	lexical	43.3%	71.7%	0.5809	81.7%
gemini_2	semantic	85.0%	100.0%	0.9167	96.7%
gemini_2	hybrid	80.0%	95.0%	0.8677	99.2%

Persian and English Results

Language	Arm	Hybrid Top-1	Hybrid Top-3	Hybrid MRR@10
FA	nomic	56.7%	76.7%	0.6778
FA	gemini_001	70.0%	96.7%	0.8222
FA	gemini_2	76.7%	96.7%	0.8556
EN	nomic	70.0%	80.0%	0.7867
EN	gemini_001	73.3%	93.3%	0.8389
EN	gemini_2	83.3%	93.3%	0.8798

Controlled 18-Case Retrieval

Method	Arm	Hit@1	Hit@3	Hit@10	MRR@10
lexical	nomic	66.7%	72.2%	72.2%	0.6944
lexical	gemini_001	66.7%	72.2%	72.2%	0.6944
lexical	gemini_2	66.7%	72.2%	72.2%	0.6944
semantic	nomic	44.4%	44.4%	44.4%	0.4444
semantic	gemini_001	55.6%	94.4%	100.0%	0.7222
semantic	gemini_2	55.6%	94.4%	100.0%	0.7315
hybrid	nomic	44.4%	61.1%	77.8%	0.5509
hybrid	gemini_001	66.7%	83.3%	100.0%	0.7677
hybrid	gemini_2	61.1%	83.3%	100.0%	0.7395

Fixed-GLM QA Diagnostic

The six-case GLM run is diagnostic only: completion does not imply answer correctness. All six prompts were byte-identical across embedding arms.

Embedding arm	Complete	Failed	Stop	Length	Mean latency
nomic	6/6	0/6	5	1	6.834s
gemini_001	6/6	0/6	5	1	7.102s
gemini_2	6/6	0/6	5	1	7.351s

Conclusion and Limitations

Strict superiority requires both official Hybrid Top-3 and MRR@10 to increase.

A ranking preference requires Top-3 and evidence recall not to regress while Top-1 and MRR@10 improve.

Official Hybrid Top-1 delta (Gemini 2 - Gemini 001): +8.3%

Official Hybrid Top-3 delta: +0.0%

Official Hybrid MRR@10 delta: +0.0371

Official Hybrid Evidence Recall@10 delta: +4.2%

Results are descriptive and limited to the frozen Python repositories and bilingual benchmark. The smaller 18-case set showed a modest Hybrid Top-1 and MRR regression for Gemini 2, so universal superiority is not claimed. Prior arms were executed at earlier dates, so latency is not used for provider ranking. No production configuration was changed.